

COMPUTER ARCHITECTURE

ADDER CIRCUIT

An adder is a device that will add together two bits and give the result as the output. The bits being added together are called the "addends". Adders can be concatenated to add two binary numbers of an arbitrary length.

There are two kinds of adders - half adders and full adders. A half adder adds two bits together and gives a two-bit output. A full adder adds two inputs and a carried input from another, giving a two-bit output.

Half Adders[

When adding two separate bits together, there are four possible combinations. Each of these is shown below with its solution.

$$\begin{array}{r} 0 \quad 1 \quad 0 \quad 1 \\ +0 \quad +0 \quad +1 \quad +1 \\ \hline 00 \quad 01 \quad 01 \quad \textcolor{red}{1}0 \end{array}$$

carried bit 

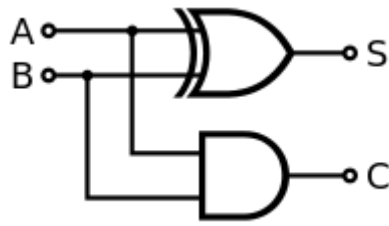
It can easily be seen that the bit in the right-hand column (the "ones" column) is a 1 only when the addends are different. XORing the addends together can therefore give us the right-hand bit. This bit is called the **sum** and is the **modulo-2 sum** of the addends (i.e., the solution if you loop around to zero again once you pass one).

The left-hand bit reads 1 only when both addends are 1, so an AND gate can be used to generate this bit, called the **carried bit**. As a summary:

$$\begin{aligned} S &= A \oplus B \\ C &= AB \end{aligned}$$

The diagram below shows the complete half adder with the addends represented by A and B, the sum represented by S, and the carried bit represented by C.

The truth table is as equally given below.ollows



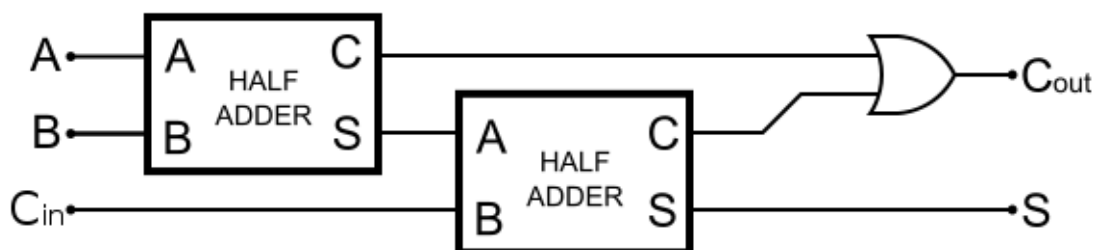
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Full Adders

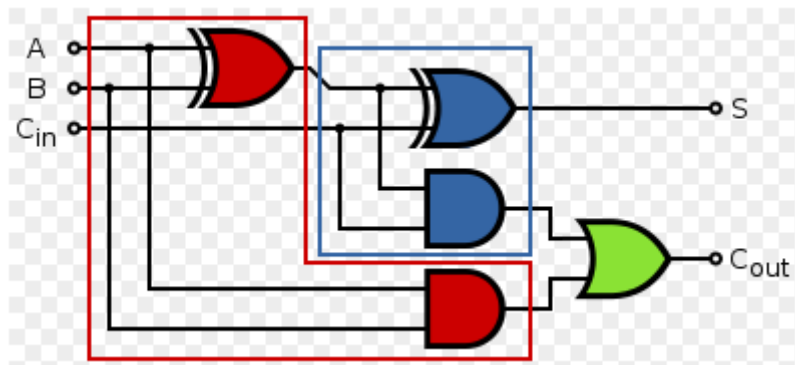
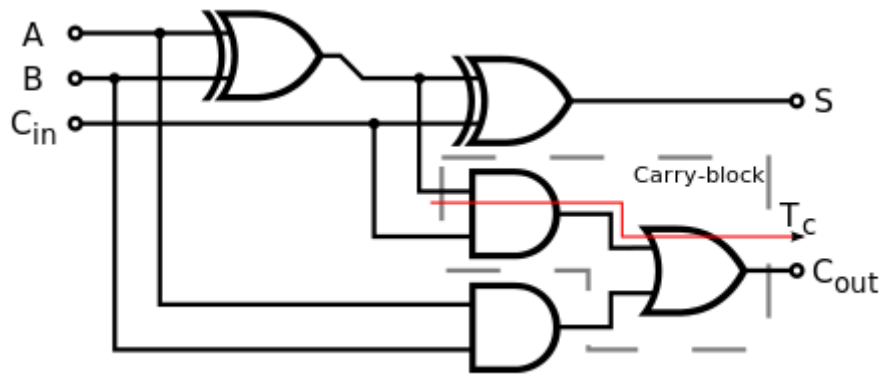
The downfall of half adders is that while they can generate a carry-out output, they cannot deal with a carry-in signal. This means that they can only ever be stand-alone units and cannot add multiple-bit numbers.

A full adder solves this problem by adding three numbers together - the two addends, as in the half adder, and a carry-in input.

A full adder can be constructed from two half adders by connecting A and B to the input of one-half adder, connecting the sum from that to an input to the second adder, connecting the carry-in, C_{in} , to the other input, and ORing the two half adder carry outputs to give the final carry output, C_{out} .



The diagrams below shows a full adder at the gate level.



The output of the full adder is the two-bit arithmetic sum of three one-bit numbers.

The logic expressions for this full adder are:

$$S = (A \oplus B) \oplus C_{in}$$

$$C_{out} = (AB) + (C_{in} (A \oplus B))$$

statement simplifies to

$$C_{out} = (AB) + (BC_{in}) + (AC_{in})$$

By applications of bool

Use truth table to prove this expression

Reference:

https://en.wikibooks.org/wiki/Practical_Electronics/Adders